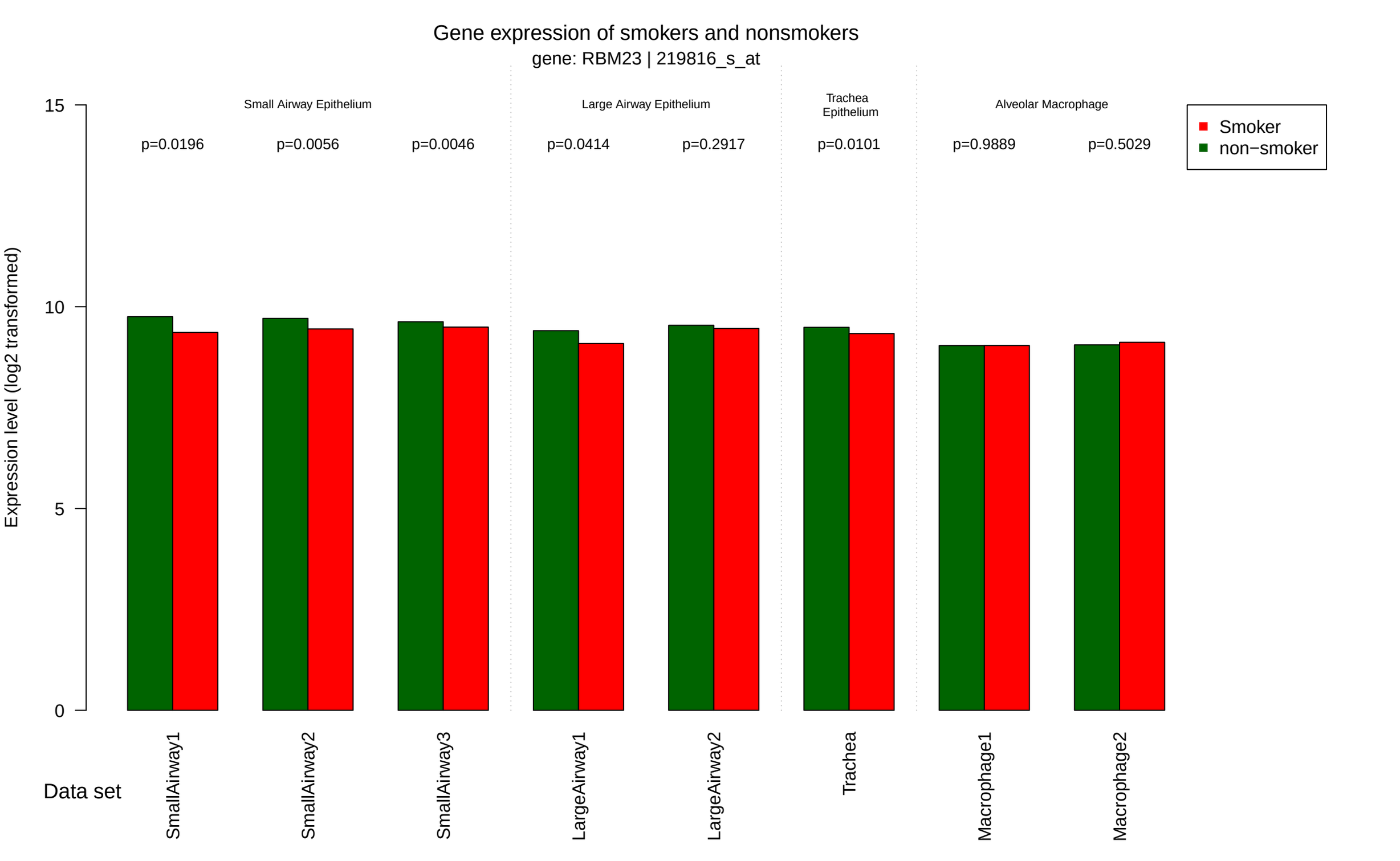
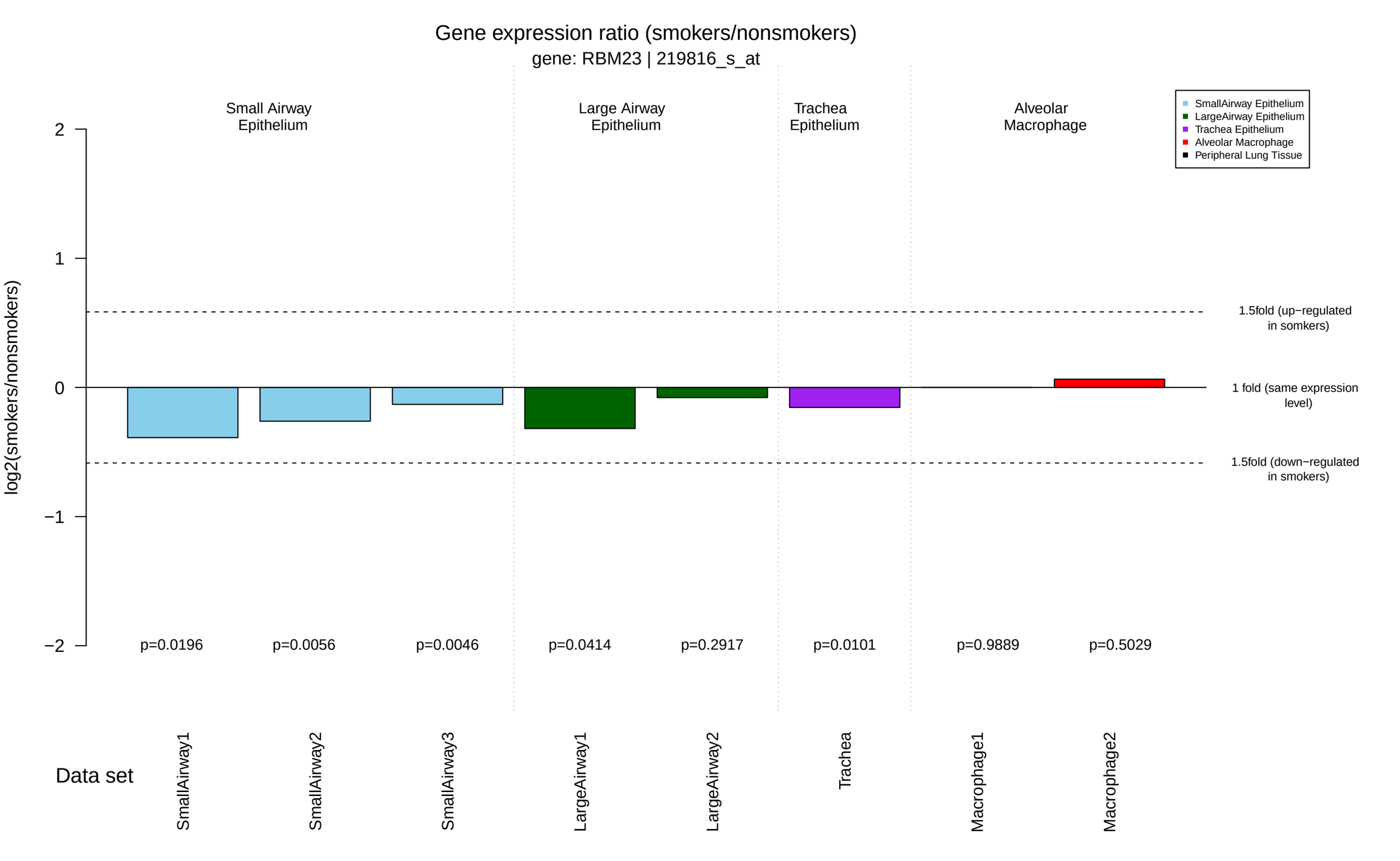
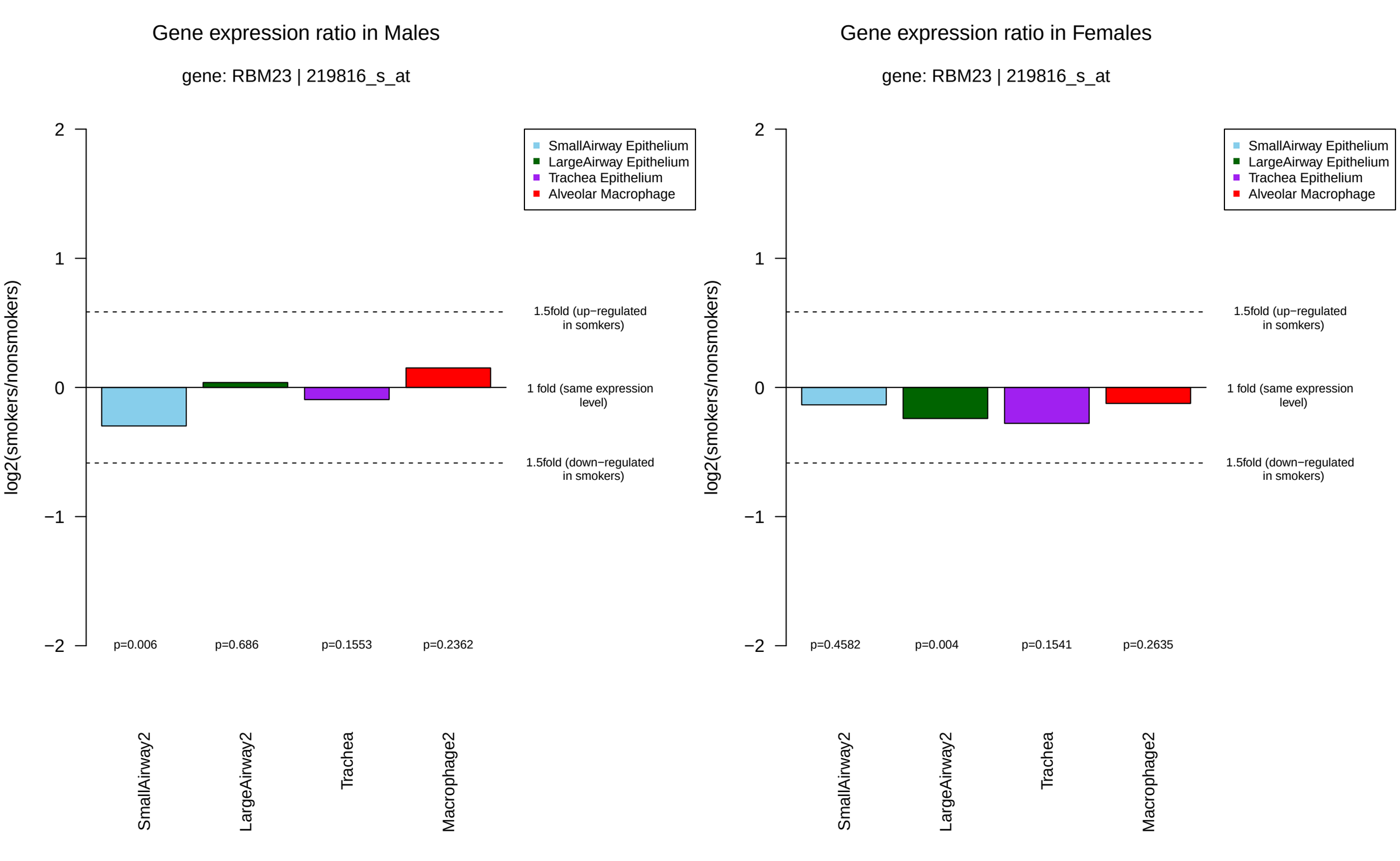
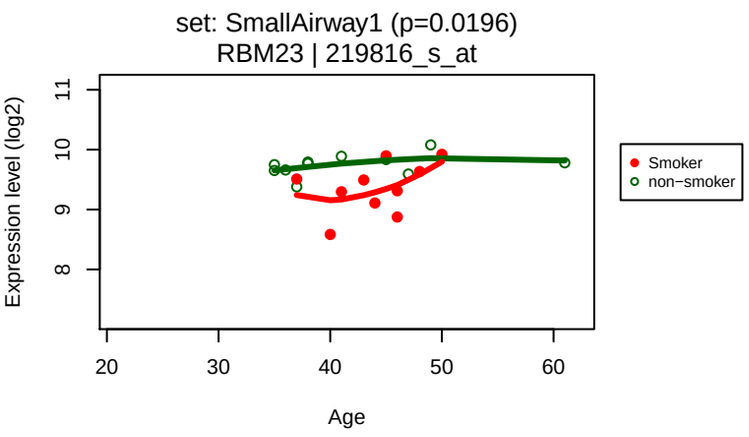
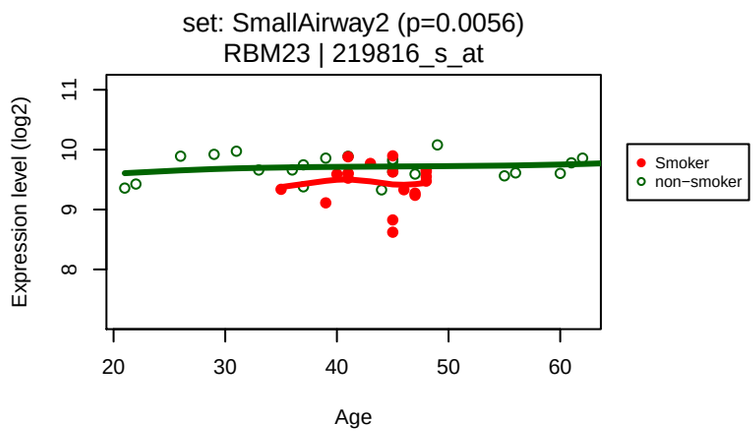
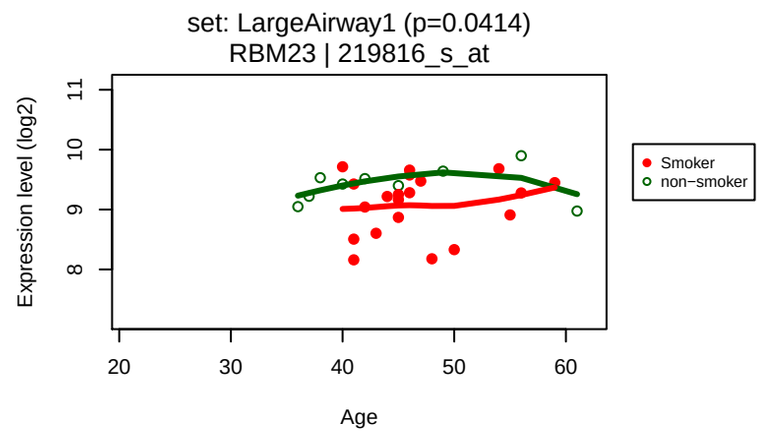

Gene expression of smokers and nonsmokers
gene: RBM23 | 219816_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0196
p=0.0056
p=0.0046
p=0.0414
p=0.2917
p=0.0101
p=0.9889
p=0.5029
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

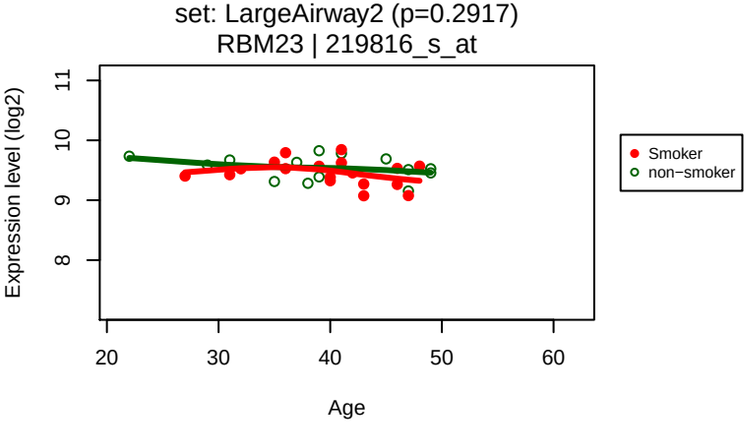

Gene expression ratio (smokers/nonsmokers)
gene: RBM23 | 219816_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0196
p=0.0056
p=0.0046
p=0.0414
p=0.2917
p=0.0101
p=0.9889
p=0.5029
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

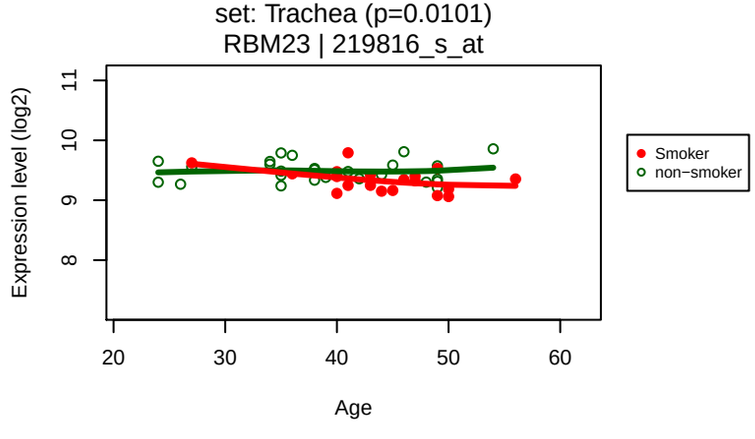

Gene expression ratio in Males
gene: RBM23 | 219816_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.006
p=0.686
p=0.1553
p=0.2362
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM23 | 219816_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4582
p=0.004
p=0.1541
p=0.2635
SmallAirway2
LargeAirway2
Trachea
Macrophage2

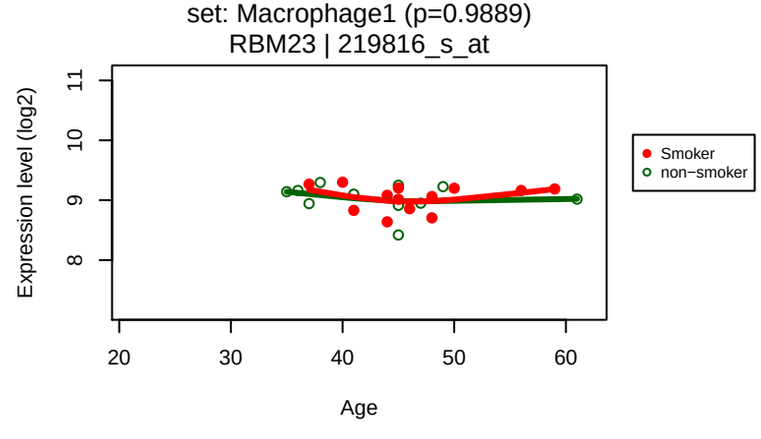

set: SmallAirway1 (p=0.0196)
RBM23 | 219816_s_at
Expression level (log2)
Age
Smoker
non−smoker

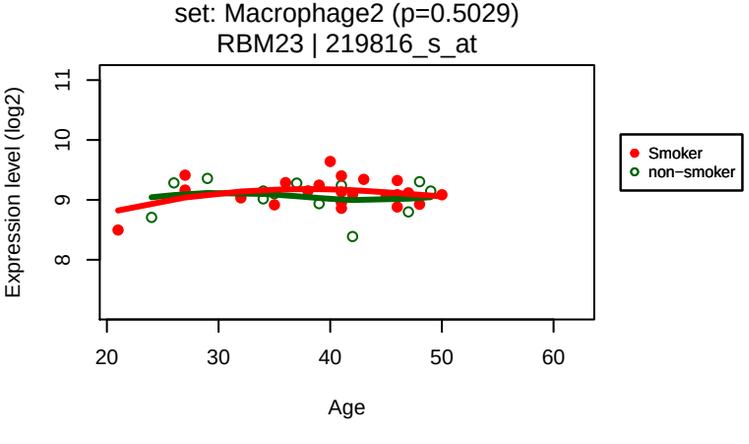

set: SmallAirway2 (p=0.0056)
RBM23 | 219816_s_at
Expression level (log2)
Age
Smoker
non−smoker

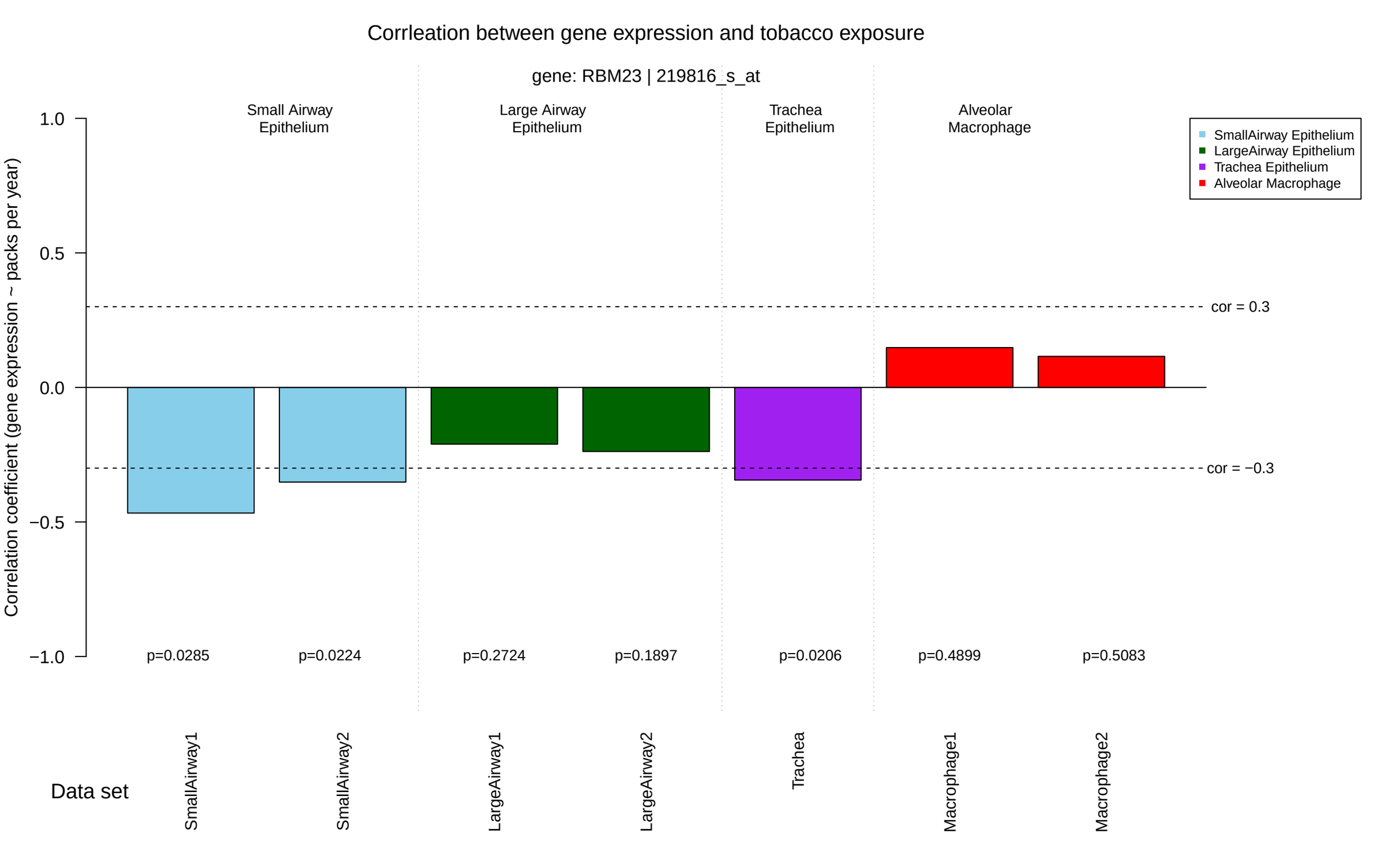

set: LargeAirway1 (p=0.0414)
RBM23 | 219816_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2917)
RBM23 | 219816_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0101)
RBM23 | 219816_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9889)
RBM23 | 219816_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5029)
RBM23 | 219816_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM23 | 219816_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0285
p=0.0224
p=0.2724
p=0.1897
p=0.0206
p=0.4899
p=0.5083
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2